Ps 01

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1 Problem 1 DONE

1.1 Task

Let Ω be any nonempty set. Define the function $d_\Omega : \Omega \times \Omega \to \mathbb{R}$ by

$$d_\Omega(x, y) = \begin{cases} 0, & \text{if } x = y, \\ 1, & \text{if } x \neq y. \end{cases}$$

(1) Show that d_Ω is a metric on Ω. (2) Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in Ω. Show that $(x_n)_{n\in\mathbb{N}}$ converges in (Ω, d_Ω) if and only if it is eventually constant. (... properly define what we mean by "eventually constant"...)

1.2 Solution

Definition (metric space). Given a set $\Omega \neq \emptyset$, we say that a mapping $d_\Omega : \Omega \times \Omega \to \mathbb{R}$ is a metric if the following three properties are satisfied:

- (M1) Symmetry: $\forall\, x, y \in \Omega$,

$$d_\Omega(x, y) = d_\Omega(y, x)$$

- (M2) Triangle Inequality: $\forall\, x, y, z \in \Omega$,

$$d_\Omega(x, z) \leq d_\Omega(x, y) + d_\Omega(y, z)$$

- (M3) Definiteness: $\forall\, x, y \in \Omega$,

$$d_\Omega(x, y) = 0 \iff x = y$$

If d_Ω is a metric on Ω, then (Ω, d_Ω) is called a metric space.

- Part (1)

  For d_Ω to be a metric on set Ω, it must satisfy three requirements of metric space.

  By the outlined definition of a metric space, I validate (Ω, d_Ω) to be a metric space.

  - (M1) Symmetry: $\forall\, x, y \in \Omega$

    Let $x, y \in \Omega$. Hence,

    * Case $x = y$

    $$d_\Omega(x, y) = 0 \wedge d_\Omega(y, x) = 0 \implies d_\Omega(x, y) = d_\Omega(y, x)$$

    Case holds.

* Case $x \neq y$

$$d_\Omega(x,y) = 1 \wedge d_\Omega(y,x) = 1 \implies d_\Omega(x,y) = d_\Omega(y,x)$$

Case holds.

Hence. requirement holds.

- (M2) Triangle Inequality: $\forall\, x, y, z \in \Omega$

  Let $x, y, z \in \Omega$. Hence,

  * Case $x = z$

$$d_\Omega(x,z) \leq d_\Omega(x,y) + d_\Omega(y,z)$$

LHS: $x = z \implies d_\Omega(x,z) = 0$ by definition.
RHS: Since metric is non-negative (d_Ω is either 0 or 1), RHS ≥ 0.

$$0 \leq d_\Omega(x,y) + d_\Omega(y,z)$$

  * Case $x \neq z$

$$d_\Omega(x,z) \leq d_\Omega(x,y) + d_\Omega(y,z)$$

LHS: $x \neq z \implies d_\Omega(x,z) = 1$ by definition.
RHS: Since $x \neq z$ this leads to 2 cases:

  · y equal to either x or z:
  RHS: $d_\Omega(x,y) + d_\Omega(y,z) = 1$
  Hence,

$$\begin{aligned} d_\Omega(x,z) &\leq d_\Omega(x,y) + d_\Omega(y,z) \\ 1 &\leq 1 \end{aligned}$$

  · y equal to neither x and z:
  RHS: $d_\Omega(x,y) + d_\Omega(y,z) = 2$
  Hence,

$$\begin{aligned} d_\Omega(x,z) &\leq d_\Omega(x,y) + d_\Omega(y,z) \\ 1 &\leq 1 + 1 \\ 1 &\leq 2 \end{aligned}$$

Hence. requirement holds.

- (M3) Definiteness: $\forall\, x, y \in \Omega$

  * $d_\Omega(x,y) = 0 \implies x = y$ by definition of a function d_Ω.
  * $x = y \implies d_\Omega(x,y) = 0$ by definition of a function d_Ω.

Hence. requirement holds.

Since the mapping $d_\Omega : \Omega \times \Omega \to \mathbb{R}$ satisfies all three constraints, d_Ω is a metric of set Ω.

- Part (2)
  - $\Rightarrow$ Convergence implies eventually constant
    - **Assumption**: Assume $(x_n)_{n\in\mathbb{N}}$ converges to x.
      - This means, $\forall\epsilon > 0, \exists N \in \mathbb{N} : \forall n > N, d(x_n, x) < \epsilon$.
    - **Goal**: Show that this implies $x_n = x$.
    - **Proof**:
      - Let $\epsilon = 0.5$.
      - Convergence implies $d_\Omega(x_n, x) < \epsilon$.
      - Since discrete distance d_Ω is binary (either 0 or 1), $d_\Omega(x_n, x) = 0$.
      - As $d_\Omega(x, y) = 0 \Leftarrow x = y$, hence $x_n = x \, \forall n > N$.
    - **Conclusion**: By definition, $(x_n)_{n\in\mathbb{N}}$ becomes eventually constant.
  - $\Leftarrow$ Eventually constant implies convergence
    - **Assumption**: Assume a sequence $(x_n)_{n\in\mathbb{N}}$ becomes eventually constant at some value x.
      - This means, $\exists N \in \mathbb{N} : \forall n > N, d_\Omega(x_n, x) = 0$
    - **Goal**: Show that this satisfies the definition of convergence $\forall\epsilon > 0$.
    - **Proof**:
      - Let $\epsilon > 0$.
      - $\forall n > N$ we know $x_n = x$.
      - Hence, $d_\Omega(x_n, x) = 0$.
      - Thus $d_\Omega(x_n, x) < \epsilon \, \forall n > N$.
    - **Conclusion**: By definition, (x_n) converges in x.

2 Problem 2 DONE

2.1 Task

Consider $\mathbb{R}^2$ with the metric

$$d(x, y) = |x_1 - y_1| + |x_2 - y_2|, \quad x = (x_1, x_2)^T, y = (y_1, y_2)^T \in \mathbb{R}^2.$$

Consider the sequences $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}^2$

$$a_n = \begin{pmatrix} \frac{1}{n+1} \\ 2 + \frac{n}{n^2+1} \end{pmatrix} \qquad b_n = \begin{pmatrix} \sqrt{4+n} - \sqrt{n} \\ (-1)^n \end{pmatrix}$$

and determine their behavior in regards to convergence in $(\mathbb{R}^2, d)$? In addition, investigate/illustrate also in Julia.

2.2 Solution

- Part $(a_n)_{n\in\mathbb{N}}$

$$\text{As } \lim_{n\to\infty} a_n^{(1)} = \lim_{n\to\infty} \frac{1}{n+1} = 0$$
$$\text{As } \lim_{n\to\infty} a_n^{(2)} = \lim_{n\to\infty} 2 + \frac{n}{n^2+1} = \lim_{n\to\infty} 2 + \frac{1}{n+\frac{1}{n}} = 2$$

Hence a candidate limit for a sequence $(a_n)_{n\in\mathbb{N}}$ is $x_l = (0,2)^T$.
Goal: Demonstrate that $\lim_{n\to\infty} d(x_n, x_l) = 0$.
Thus,

$$\begin{aligned}
\lim_{n\to\infty} d(x_n, x_l) &= \lim_{n\to\infty} \left|\frac{1}{n+1} - 0\right| + \left|2 + \frac{n}{n^2+1} - 2\right| \\
&= \lim_{n\to\infty} \left|\frac{1}{n+1}\right| + \left|\frac{n}{n^2+1}\right| \\
&= \lim_{n\to\infty} \left|\frac{1}{n+1}\right| + \lim_{n\to\infty} \left|\frac{n}{n^2+1}\right| \\
&= 0 + \lim_{n\to\infty} \left|\frac{n}{n^2+1}\right| \\
&= 0 + \lim_{n\to\infty} \left|\frac{1}{n+\frac{1}{n}}\right| \\
&= 0 + 0 \\
&= 0
\end{aligned}$$

Hence, $x_l = (0,2)^T$ is indeed a limit of a sequence $(a_n)_{n\in\mathbb{N}}$.

- Part $(b_n)_{n\in\mathbb{N}}$

$$\begin{aligned}
\text{As } \lim_{n\to\infty} b_n^{(1)} &= \lim_{n\to\infty} \sqrt{4+n} - \sqrt{n} \\
&= \lim_{n\to\infty} \frac{(\sqrt{4+n} - \sqrt{n})(\sqrt{4+n} + \sqrt{n})}{(\sqrt{4+n} + \sqrt{n})} \\
&= \lim_{n\to\infty} \frac{(4+n) - (n)}{\sqrt{4+n} + \sqrt{n}} \\
&= \lim_{n\to\infty} \frac{4}{\sqrt{4+n} + \sqrt{n}} \\
&= 0 \\
\text{As } \lim_{n\to\infty} b_n^{(2)} &= (-1)^n
\end{aligned}$$

$b_n^{(2)}$ would alternate between the values of -1 and 1, since $n \in \mathbb{N}$ is either even or odd. Since the second component diverges, the sequence $(b_n)_{n\in\mathbb{N}}$ can not converge in $(\mathbb{R}^2, d)$.

- Code

```
using Plots

a_n1(n) = 1 / (n + 1)
a_n2(n) = 2 + (n / (n^2 + 1))
b_n1(n) = sqrt(4+n) - sqrt(n)
b_n2(n) = (-1)^n

d(x,y) = abs(x[1] - y[1]) + abs(x[2] - y[2])

n = 1:100

a_x = a_n1.(n)
a_y = a_n2.(n)

b_x = b_n1.(n)
b_y = b_n2.(n)

p1 = scatter(a_x, a_y,
             title="sequence a_n",
             label="a_n points",
             xlabel="x1", ylabel="x2",
             legend=:bottomright)

p2 = scatter(b_x, b_y,
             title="sequence b_n",
             label="b_n points",
             xlabel="x1", ylabel="x2",
             legend=:bottomright)

plot(p1, p2, layout=(2,1), size=(1200, 1200))

savefig("./figs/p_02.png")
```

```
/Users/ziro2173/personal/university/01_bachelor/2025_winter_term/01_analysis/02_exercises/s
```

3 Problem 3 DONE

3.1 Task

Consider the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}.$$

(1) Show that it converges towards a limit $a \in \mathbb{R}$ Does it converge absolutely? (2) Use Julia to guess the limit of $\sum_{k=0}^{\infty} \frac{a^k}{k!}$

3.2 Solution

- Part (1)

  Observe the product $(-1)^{n+1} \cdot \frac{1}{n}$. As $n \to \infty$, $(-1)^{n+1}$ would alternate between the values of -1 and 1. Hence the series is alternating.

  First, I will evaluate whether the alternating series converge by using the Alternating Series Test.

  - First condition of AST is a series' sequence being equal to 0 at $n \to \infty$.

    $$\lim_{n\to\infty} \frac{1}{n} = 0$$

    Condition met.
  - Second condition of AST is series' sequence next term being less than preceding one $a_{n+1} < a_n$.

    $$\frac{1}{n+1} < \frac{1}{n}$$

    Condition met.

  Since both conditions are met, by the Alternating Series Test, the series converges to some limit $a \in \mathbb{R}$.

  Secondly, I have to validate whether the series converges absolutely.

$$
\begin{aligned}
\sum_{n=1}^{\infty}\left|\frac{(-1)^{n+1}}{n}\right| &= \sum_{n=1}^{\infty}\frac{|(-1)^{n+1}|}{|n|} \\
&= \sum_{n=1}^{\infty}\frac{|(-1)^{n+1}|}{n} \quad \text{(since } n \in \mathbb{N}\text{)} \\
&= \sum_{n=1}^{\infty}\frac{1}{n} \\
&= \frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+...
\end{aligned}
$$

This is a harmonic series that is known to be divergent. Hence, series do not converge absolutely.

- Part (2)

```
s_conv() = sum((-1)^(n+1)/n for n in 1:100_000)
a = s_conv()

n_conv() = sum((a^k)/factorial(k) for k in 0:20)

println("Value of a: ", a)
println("Value of Sum: ", n_conv())
```

```
s_conv (generic function with 1 method)
0.6931421805849816
n_conv (generic function with 1 method)
Value of a: 0.6931421805849816
Value of Sum: 1.999990000075072
```

4 Problem 4

4.1 Task

We want to show that the sequence $(a_n)_{n\in\mathbb{N}} = ((1+\frac{1}{n+1})^{n+1})_{n\in\mathbb{N}}$ contains a convergent subsequence by verifying the following steps:

(1) Show that $(a_n)_{n\in\mathbb{N}}$ is bounded. Hint for the upper bound: use the binomial theorem to verify

$$(1+\frac{1}{n})^n \leq \sum_{k=0}^{n}\frac{1}{k!}, \quad n = 1, 2, ...$$

(2) Argue now that $(a_n)_{n\in\mathbb{N}}$ has a convergent subsequence. Make additionally numerical checks in Julia to collect some evidence that $(a_n)_{n\in\mathbb{N}}$ is monoton and hence convergent.

4.2 Solution

- Part (1)

$$(a+b)^n = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \cdot a^{n-k} b^k$$

- Part (2)

```
nil
```